

NAGALAND UNIVERSITY

(A Central University Established by an Act of Parliament 1989)

SCHOOL OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF INFORMATION TECHNOLOGY

Kohima Campus, Nagaland – 797004



Assignment 2

ITB301: Object Oriented Programming

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Semester: 3rd Sem

Submitted To:

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Date: September 25, 2026

1. Write a Program to calculate the integration of $\int_0^2 x^4 dx$ using Simpson's Rule.

Code:

```
1  #include <cmath>
2  #include <iomanip>
3  #include <iostream>
4
5  using namespace std;
6
7  class Integrate {
8      double fx(double x) { return pow(x, 4); }
9      // f(x) = x^4
10
11     public:
12         int n;
13         double a = 0, b = 2;
14         // lower bound: 0, upper bound: 2
15
16         double integrate() {
17             double h = (b - a) / n;
18             double sum = 0;
19             int mul;
20             for (int i = 0; i <= n; i++) {
21                 if (i == 0 || i == n) {
22                     mul = 1;
23                 } else if (i % 2 == 0) {
24                     mul = 2;
25                 } else {
26                     mul = 4;
27                 }
28                 sum += mul * fx(i * h);
29             }
30             return sum * (h / 3);
31         }
32     };
33
34     int main() {
35         Integrate i;
36         cout << "Enter the number of iterations: ";
37         cin >> i.n;
38         if (i.n % 2 != 0) {
```

C++

```
39         cout << "Input must be an even number" << endl;
40         return 1;
41     }
42     cout << "The result is: " << fixed << setprecision(15) <<
43         i.integrate()
44         << endl;
45     return 0;
46 }
```

Output:

```
1 Enter the number of iterations: 10
2 The result is: 6.400426666666667
```

```
1 Enter the number of iterations: 7500
2 The result is: 6.400000000000006
```

```
1 Enter the number of iterations: 9999
2 Input must be an even number
```